Tenko on CICIoT2023: An End-to-End Experiment Report

A discussion memo for internal circulation

Tenko team

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Abstract

This memo documents, end to end, our evaluation of Tenko on **CICIoT2023**, the recent IoT dataset requested by Reviewers 1.2 and 3.2. It explains *why* each decision was made: why we used the raw PCAP rather than the CSVs (Tenko's node key is the per-packet source IP), how we built the benign/attack streams, and the two changes we made to run Tenko on this dataset — a benign-only recalibration of the tolerance factors η (the committed values are degenerate here) and the handling of a redundant double-tanh in the committed pipeline. We headline threshold-independent **AUC/EER** because we found that a fixed benign-calibrated η does *not* reproduce a target false-positive rate out-of-sample on this non-stationary benign trace. We then compare against the paper's existing Kitsune-dataset results. Bottom line: CICIoT2023 is **consistent with the paper's reliable-vs-degraded story**, Tenko's node-aggregation advantage reproduces on the sustained-deviation attacks (MITM, Mirai), but Tenko is *competitive, not uniformly superior* to Kitsune here — a **qualified yes**. All figures are traceable (Appendix A).

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1 Motivation: why CICIoT2023, and why now

The R1 reviewers asked us to justify evaluating on datasets that are several years old and to demonstrate the method on a *recent* trace. Specifically, Reviewer 1.2 ("run recent, CIC IDS 2023") and Reviewer 3.2 ("motivate old datasets; recency") both request a modern dataset. There is no separate "CIC-IDS2023"; the reviewers' request maps to **CICIoT2023** (University of New Brunswick / Canadian Institute for Cybersecurity, `https://www.unb.ca/cic/datasets/iotdataset-2023.html`).

CICIoT2023 is the *right* recent dataset for Tenko for three concrete reasons:

1. It ships **raw PCAP**, so we get *ordered per-packet streams* — Tenko is a streaming, packet-level detector.
2. It provides a **benign reference window**, which Tenko needs for benign-only calibration (no attack labels at training time).
3. It preserves **per-packet source identity** (source IP), which is Tenko's *node key* for the node-level aggregation that is the paper's core contribution.

These are exactly the properties several other "recent" options lack (see Section 2).

2 The dataset

What it is. CICIoT2023 is a large IoT security dataset built from a testbed of **105 real IoT devices**, containing benign traffic and **7 attack families** (DDoS, DoS, Recon, Web, Brute Force, Spoofing, and Mirai). The full raw PCAP corpus is very large (≈548 GB); the CSV package alone is ≈13 GB.

The load-bearing decision: PCAP, not CSV. Tenko's contribution is node-level aggregation, and a node is identified by the **per-packet source IP**. The widely used CICIoT2023 *CSVs drop that column* — we verified the downloaded CSV package is "39 aggregate features + Label with **no** source IP / flow-id / timestamp," which makes node aggregation impossible (not even a pseudo-node). The same disqualifier applies to the earlier "IDS 2025" folder we investigated, which turned out to be **CSE-CIC-IDS2018** flow CSVs (80 CICFlowMeter features + `Label`, *no* source-IP column) — incompatible with Tenko's packet pipeline. We therefore ingest CICIoT2023 from **PCAP**.

Classes used and raw sizes. We used a **6-class pilot**, one PCAP per class, plus benign traffic (4 chunks, ≈7 GB from `BenignTraffic*.pcap`). Table 1 lists the classes and their raw on-disk sizes.

Label semantics. CICIoT2023 has **no per-packet ground-truth file**; the class *is* the capture-session identity — i.e., the PCAP a packet came from. The binary label is 0 if the packet is from `BenignTraffic`, and 1 otherwise. This session-level labelling is standard for this dataset and is why we build one benign-vs-single-attack stream per class.

Table 1: The six CICIoT2023 attack classes used in the pilot and their raw PCAP sizes on disk.

Class	**Source PCAP**	**Raw size**
DDoS-UDP Flood	`DDoS-UDP_Flood.pcap`	2.048 GB
DoS-SYN Flood	`DoS-SYN_Flood.pcap`	2.048 GB
Recon (OS Scan)	`Recon-OSScan.pcap`	324 MB
MITM (ARP Spoofing)	`MITM-ArpSpoofing.pcap`	2.046 GB
Mirai (greeth flood)	`Mirai-greeth_flood.pcap`	2.048 GB
Dictionary Brute Force	`DictionaryBruteForce.pcap`	39 MB

Table 2: Per-attack stream layout (identical for all six classes). `benignLimit` = length of benign_lead = 150,000.

Region	**Label**	**Index range (0-based)**	**Size**
benign_lead (KitNET train + pattern calibration)	0	0 . . . 149,999	150,000
benign_test (FPR negatives)	0	150,000 . . . 179,999	30,000
attack (TPR positives)	1	180,000 . . . 229,999	50,000
total per stream	—	—	**230,000**

3 Experimental protocol

PCAP → TSV. Every PCAP is converted with `tshark` to the exact 19-column field order Tenko's `FeatureExtractor` expects. The **source IP** — the node key — is **column 4 (IPv4)** or **column 17 (IPv6)**; the conversion script warns "do not reorder," and the order was verified against `FeatureExtractor.pcap2tsv_with_tshark()`.

Pilot subsampling ("first-N"). Because the class PCAPs are multi-GB, we extract the *first N packets* per source with `tshark -c N -w` (it stops early instead of scanning the whole file). The window sizes are fixed constants: $N_{\text{lead}} = 150{,}000$, $N_{\text{test}} = 30{,}000$, $N_{\text{atk}} = 50{,}000$.

Stream layout (Table 2). Each per-attack stream is the concatenation [**benign_lead**] + [**benign_test**] + [**attack**] with packet order preserved, giving 230,000 packets, identical for all six classes.

Why benign train and test come from the *same* trace. `benign_lead` is the first 150,000 packets of `BenignTraffic.pcap`; `benign_test` is packets 150,001 . . . 180,000 of the *same* file. Drawing both from one benign trace follows the standard Kitsune protocol (benign train + benign test from a single trace) and avoids cross-session distribution shift that would artificially inflate the measured FPR. The attack region is the first 50,000 packets of the class PCAP.

KitNET grace. Within KitNET, the first FMgrace = 5,000 (feature-mapping) + ADgrace = 50,000 (anomaly-detector) packets are grace/training, so benign *calibration* for thresholds begins at index 55,001 and runs to `benignLimit` = 150,000.

What the driver runs. For each stream, `run_ciciot2023.py`:

Table 3: Committed vs benign-recalibrated tolerance factors. Recalibrated values are identical across classes because the benign_test spread is shared. The fix is overwhelmingly on η_{node} ($20 \rightarrow\sim$ 2.2–2.85), confirming it was mis-scaled.

Setting	η_{global}	η_{node}	**shared** q
committed (degenerate here)	50.000	20.000	—
recalibrated, target benign FPR = 1%	40.699	2.854	0.994768
recalibrated, target benign FPR = 5%	35.834	2.207	0.968434

1. Streams the TSV through **KitNET (X1)** to produce one raw RMSE per packet, with a guaranteed 1:1 RMSE↔packet↔label alignment (a parse error yields a neutral 0.0 rather than truncating the stream).

2. Runs the **Kitsune baseline**: per-packet tanh(RMSE) with a benign Median+MAD threshold, $\text{thr} = \text{median} + 3.0 \cdot 1.4826 \cdot \text{MAD}$, calibrated on [55,001 : 150,000] and evaluated on the test region.

3. Runs **Tenko X2–X4** via `get_adversarial_IPs_weighted_pattern(·)` with `benignLimit`= 150,000, `memorySize`= 60, `pattern_window_size`= 100, `pattern_segments`= 10, fusion weights 0.5/0.5, threshold 0.5.

What it persists. The driver saves per-test-packet arrays — `arr_gold_*`, `arr_ndg_*` and `arr_nds_*` (the global and single-aggregate *normalized* pattern distances), `arr_cont_*` (fused distance), `arr_node_*` (node score), and `arr_kitsune_*` (Kitsune score). Persisting these lets us re-search η and compare the tanh variants *offline*, without re-running X1/X2.

4 Changes we made, and why

4.1 Benign-only η recalibration (the main change)

Symptom. With the committed operating point $\eta_{\text{global}} = 50$, $\eta_{\text{node}} = 20$ (fusion 0.5/0.5, threshold 0.5), Tenko was **degenerate on all six classes — it flagged nothing** (TPR = FPR = 0).

Root cause. The fused rule fires when a *normalized* pattern distance $\text{nd} > \eta$, where $\text{nd} = \text{distance}/(\text{max benign training spread})$. After the committed squashing, the benign-normalized attack distances top out around ~ 2.4 — far below $\eta_{\text{node}} = 20$ and $\eta_{\text{global}} = 50$ — so **nothing ever fires**. In other words, η_{node} was mis-scaled by roughly an order of magnitude for this dataset's benign spread. This is not a harness bug; the committed constants simply do not transfer to CICIoT2023's benign geometry.

Method (benign-quantile search, no attack labels). `recalibrate_ciciot2023.py` scans a shared benign quantile q over [0.50, 0.99999], sets $\eta_{\text{global}} = Q(\text{benign\_ndg}, q)$ and $\eta_{\text{node}} = Q(\text{benign\_nds}, q)$ (the q-quantiles of the *benign* normalized distances), and selects the q whose *fused benign-only* FPR is as close as possible to — but not above — the target. Crucially, **no attack labels are used** to choose η; the values are a property of benign traffic only. Table 3 lists the result.

Table 4: Single- vs double-tanh pattern-distance AUC on CICIoT2023. The maximum change is +0.010911 (Dictionary Brute Force); AUC is essentially invariant to the redundant second tanh.

Attack	**double-**tanh **AUC**	**single-**tanh **AUC**	**Δ (single−double)**
DDoS-UDP Flood	0.928822	0.930948	+0.002126
DoS-SYN Flood	0.987792	0.988392	+0.000600
Recon (OS Scan)	0.795871	0.796219	+0.000348
MITM (ARP Spoofing)	0.949187	0.951141	+0.001954
Mirai (greeth flood)	0.936728	0.936996	+0.000268
Dictionary Brute Force	0.696816	0.707727	+0.010911

Why these values are legitimate. The paper already frames η as "a multiplier on benign deviation magnitude, *not* a label-tuned operating point" and states "attack labels are not used to select these values." Our recalibration is exactly that: a per-deployment benign multiplier, re-derived from CICIoT2023's benign traffic. We did not tune η against attack outcomes.

4.2 The double-tanh

Where. The committed pipeline applies tanh **twice**: once in `example.py:157` (`rmse = np.tanh(rmse)`, needed because node scoring requires a $[0,1]$ input) and again in `results.py:805` under a "Normalization" comment (`RMSEs_norm = np.tanh(...)`). Composed, the released code computes tanh(tanh(raw)), squashing scores into $\approx [0,\ 0.762]$. The second tanh is **redundant**: it receives an already-squashed score.

Impact on AUC is small (Table 4). Removing the redundant second tanh ("single") changes the pattern-distance AUC by at most +0.010911 (Dictionary Brute Force), with a median well under 0.002 — i.e. **AUC change** ≤ 0.011.

Impact on the operating point is larger. Under *single*-tanh the committed $\eta = 50/20$ stops being uniformly degenerate — it fires for 4/6 attacks (e.g. DoS-SYN TPR 0.71518, DDoS TPR 0.05964) — whereas under *double* it stays degenerate. This is why the second tanh matters more for thresholding than for ranking.

Reporting decision and provenance. We report CICIoT2023 under a **fixed single-**tanh normalization (redundant second tanh removed for this dataset), noting it is numerically ≈ the double variant (AUC $\leq$ 0.011). Separately, we verified that the *main-paper* per-attack numbers were themselves generated under the *double*-tanh pipeline; those published tables therefore **stand as-is** and are described accurately, not silently changed. Any change to the main paper's normalization would be coordinated with co-authors, not made unilaterally.

4.3 Out-of-sample (OOS) calibration protocol — added for honesty

The in-sample recalibration fits η on `benign_test` and then measures FPR on that *same* slice, so its 1%/5% FPR is **definitional, not predictive**. To test whether the FPR actually transfers, `recalibrate_ciciot2023_oos.py` splits `benign_test` into two disjoint halves — **CALIB = first** 15,000, **EVAL = held-out** 15,000 — selects η on CALIB only (no attack labels), and measures FPR on the disjoint EVAL benign plus the attack region. Because the benign trace has a warm-up

Table 5: Full-trace AUC/EER on CICIoT2023 (benign-only training). Higher AUC and lower EER are better; the winner column marks the better AUC (Dictionary is mixed: Kitsune AUC but Tenko EER).

Attack	**AUC Kitsune**	**AUC Tenko**	**EER Kitsune**	**EER Tenko**	**Winner**
DDoS-UDP Flood	0.965073	0.928822	0.123863	0.129363	Kitsune
DoS-SYN Flood	0.992747	0.987792	0.036643	0.051803	Kitsune
MITM (ARP Spoofing)	0.942003	**0.949187**	0.147373	**0.130690**	**Tenko**
Mirai (greeth flood)	0.929418	**0.936728**	0.153343	**0.139573**	**Tenko**
Recon (OS Scan)	0.827176	0.795871	0.216383	0.255887	Kitsune
Dictionary Brute Force	0.703386	0.696816	0.318927	0.284700	mixed

Table 6: Tenko operating-point metrics at benign-calibrated η targeting 5% FPR. FPR is in-sample (threshold set on test-benign); it is a calibration diagnostic, not a deployment FPR.

Attack	**TPR**	**FPR (in-sample)**	**Precision**	**F1**
DoS-SYN Flood	0.962660	0.049967	0.969798	0.966216
MITM (ARP Spoofing)	0.824020	0.049967	0.964895	0.888910
DDoS-UDP Flood	0.783180	0.049967	0.963131	0.863884
Mirai (greeth flood)	0.711620	0.049967	0.959574	0.817203
— reliability divider (below: genuinely low AUC, a separability limit) —				
Recon (OS Scan)	0.114620	0.049967	0.792669	0.200280
Dictionary Brute Force	0.030540	0.049967	0.504627	0.057594

transient (it is non-stationary), we report **both split directions**: `fwd` (calibrate on the first half) and `rev` (calibrate on the second half). The result is Section 5.3.

5 Results

5.1 Full-trace threshold-independent AUC / EER (headline)

Table 5 reports AUC and EER over the full held-out test trace (benign_test + attack) for Kitsune and Tenko. Tenko's AUC/EER come from the continuous η-normalized fused pattern distance and are **invariant to the operating point**. Tenko **wins on MITM and Mirai** — the sustained-deviation attacks its node aggregation targets — and is competitive but not dominant elsewhere.

5.2 Operating-point metrics at benign-calibrated η (in-sample)

Table 6 gives Tenko's operating point at the 5% benign-FPR calibration (Table 3). **The FPR column is in-sample by construction** (the tolerance is fit and the FPR measured on the same benign_test slice); it is a calibration diagnostic, *not* a deployment guarantee (see Section 5.3). The reliability divider follows the paper's convention. For reference, the Kitsune baseline runs at a **constant** benign Median+MAD FPR $\approx$ 0.3233 on every class — i.e. it is *not* operating at a comparably tight false-alarm budget, so these two systems are not compared at a matched FPR.

Table 7: Out-of-sample held-out benign FPR (double variant). `fwd`: calibrate on the first benign half, evaluate on the second; `rev`: the reverse. The realized FPR ranges from $\approx 0\%$ to $\approx 30\%$ against a 1–5% target.

Target benign FPR	fwd (calib = 1st half)	rev (calib = 2nd half)
1%	0.000	0.304000
5%	0.000	0.319267

Table 8: The paper's existing Kitsune-dataset AUC/EER (Tenko is the better value on all nine attacks). Source: manuscript table `tab:kitsune-tenko-new-highlighted`.

Attack	AUC Kitsune	AUC Tenko	EER Kitsune	EER Tenko
Mirai Botnet	0.95	0.996	0.05	<0.01
OS Scan	0.90	0.96	0.10	0.04
SSDP Flood	0.88	0.99	0.12	0.01
SSL Renegotiation	0.92	0.96	0.08	0.08
ARP MiTM	0.96	0.96	0.04	0.02
SYN DoS	0.89	0.95	0.11	0.09
Video Injection	0.93	0.95	0.07	0.05
Fuzzing	0.85	0.91	0.15	0.11
Active Wiretap	0.87	0.93	0.13	0.05

5.3 The out-of-sample FPR non-transfer finding (the honest caveat)

Table 7 shows the held-out benign FPR when η is calibrated on one benign half and measured on the disjoint other half. The same recipe yields $\approx 0\%$ FPR in one direction and $\approx 30\%$ in the other, versus the 1–5% nominal target. **A fixed benign-calibrated η does not transfer** across disjoint benign slices of this non-stationary trace. This is why the in-sample 1%/5% FPR must be labelled as definitional, and why the *threshold-independent* AUC/EER (Table 5) are the honest headline.

For completeness, on the OOS held-out slice the threshold-independent numbers remain stable, e.g. DoS-SYN Tenko AUC/EER = 0.993620/0.024487 and MITM = 0.984969/0.065113; single-tanh tracks double within ± 0.02 TPR, so the tanh choice is numerically irrelevant on this data.

6 Comparison: CICIoT2023 vs the paper's Kitsune-dataset results

The paper's established result. On the Kitsune dataset the paper reports AUC/EER for nine attacks with **Tenko $\geq$ Kitsune on all nine** (Table 8). This is the baseline story we are checking CICIoT2023 against.

Matched attacks. Four attacks appear in *both* datasets. Table 9 puts Tenko's AUC side by side. The floods and ARP-spoofing are in the same ballpark (SYN DoS is actually better on CICIoT2023); the one real divergence is **OS Scan**.

Why OS Scan diverges. The Mirsky "OS Scan" is an aggressive, well-separated scan (AUC 0.96); CICIoT2023 "Recon-OSScan" reconnaissance overlaps benign traffic far more, so the same

Table 9: Matched-attack Tenko AUC: paper's Kitsune dataset vs CICIoT2023 (both full-trace, threshold-independent). Positive Δ means CICIoT2023 is better.

Attack (matched)	**AUC (Kitsune ds)**	**AUC (CICIoT2023)**	Δ
SYN DoS $\leftrightarrow$ DoS-SYN	0.95	0.988	+0.038 (better)
ARP MITM $\leftrightarrow$ MITM-Arp	0.96	0.949	−0.011 ($\approx$ equal)
Mirai $\leftrightarrow$ Mirai-greeth	0.996	0.937	−0.059 (mildly worse)
OS Scan $\leftrightarrow$ Recon-OSScan	0.96	0.796	−0.164 (worse)

name denotes materially harder traffic (AUC 0.796). This is a traffic-character difference, not a metric error, and it is consistent with the paper's thesis (benign-overlapping traffic is hard).

Verdict — qualified yes. CICIoT2023 is **consistent with the paper's reliable-vs-degraded story**:

- **Sustained-deviation attacks stay strong**: DoS-SYN (AUC 0.988), DDoS (0.929), MITM (0.949), Mirai (0.937).
- **Benign-overlapping / low-rate attacks stay weak**: Recon-OSScan (0.796) and Dictionary Brute Force (0.697) — low AUC is a genuine separability limit.
- **The node-aggregation edge reproduces** on exactly the two entity/sustained attacks it targets (MITM +0.007, Mirai +0.008 AUC over Kitsune).

The qualification: on CICIoT2023 Tenko is **not uniformly superior** to Kitsune (2/6 AUC wins here vs 9/9 on the Kitsune dataset); on pure volumetric floods a well-thresholded per-packet detector is already near-ceiling and aggregation adds little. Combined with the per-deployment η recalibration, the weaker OS-Scan/recon, the harder benign regime (uniformly higher EER), the 6-attack pilot scope, and the FPR non-transfer, the honest summary is a **qualified yes**: Tenko generalizes to a recent dataset in the way the mechanism predicts, and we should report it that way rather than as a blanket win.

6.1 Unified view: Tenko across all datasets in the paper

Table 10 consolidates *every* corpus for which the manuscript (or this pilot) carries usable Tenko numbers. Granularity and reported metrics differ by dataset — per-attack, per-device, or a single mixed-attack regime — so we surface exactly what exists and aggregate AUC **only** over the values we actually extracted (the n column states how many). Two datasets carry no AUC we could use and are therefore *excluded from the AUC aggregates*, not fabricated: **N-BaIoT** appears only as a per-device F1/TPR/FPR table that is *commented out* in the current draft (supervised Hermes vs. benign-only Tenko; no AUC/EER), and **CIC-IDS2018** has *no results table* in the manuscript at all (it is named only in setup/roadmap discussion). The one extra active Tenko number beyond the two per-attack panels is the *Kitsune combined-attack* trace under a Mateen-style sustained-distribution-shift stress (`tab:mateen_comparative_performance`), where Tenko scores a single AUC–ROC of 0.607 against fully adaptive baselines (Mateen 0.722, OWAD 0.710, INSOMNIA 0.473, reproduced from prior work).

Caveat (read before the table). AUC across datasets is *not* a controlled comparison: the traffic, thresholds, class granularity, and — for the Mateen regime — an explicit distribution-shift

Table 10: Tenko across all datasets with usable numbers in the paper (plus this CICIoT2023 pilot). AUC aggregates are mean [min–max] over the n extracted per-class/per-regime values only. This is a generalization overview across heterogeneous protocols, *not* a controlled ranking.

Dataset	**Year**	**Granularity**	n	**Tenko AUC mean [min–max]**	**Baseline AUC mean [min–max]**	**Notes / source**
Kitsune (Mirsky)	2018	per-attack	9	0.956 [0.91–0.996]	0.906 [0.85–0.96]	Active; `tab:kitsune-tenko-new-highlighted`. Baseline = Kitsune; Tenko $\geq$ Kitsune on all 9.
CICIoT2023 (this work)	2023	per-attack	6	0.883 [0.70–0.988]	0.893 [0.70–0.993]	`ciciot2023_per_class_metrics.csv`. Baseline = Kitsune; Tenko > Kitsune on MITM, Mirai.
Kitsune combined-attack (shift stress)	2018	single mixed regime	1	0.607	0.722 (Mateen)	AUC–ROC; `tab: mateen_comparative_performance`. Adaptive baselines: OWAD 0.710, INSOMNIA 0.473.
N-BaIoT	2018	per-device	9	—	—	F1/TPR/FPR only, **commented-out** in draft; Tenko F1 0.67–0.88 vs supervised Hermes. No AUC $\Rightarrow$ excluded.
CIC-IDS2018	2018	—	—	—	—	**No results table** in manuscript $\Rightarrow$ excluded.

Table 11: Matched per-attack alignment: Tenko AUC on the Kitsune dataset vs CICIoT2023 (both full-trace, threshold-independent). Positive Δ means CICIoT2023 is better. This is the only directly comparable, same-attack axis across the two per-attack datasets.

Attack (matched)	**Tenko AUC (Kitsune ds)**	**Tenko AUC (CICIoT2023)**	Δ**AUC**
SYN DoS $\leftrightarrow$ DoS-SYN	0.95	0.988	+0.038
ARP MITM $\leftrightarrow$ MITM-Arp	0.96	0.949	−0.011
Mirai $\leftrightarrow$ Mirai-greeth	0.996	0.937	−0.059
OS Scan $\leftrightarrow$ Recon-OSScan	0.96	0.796	−0.164

stress all differ. Table 10 is a *generalization overview*, not a ranking; the only genuinely matched, like-for-like axis is the per-attack alignment in Table 11.

The matched per-attack alignment (the fair axis) is repeated here for convenience (Table 11): four attacks recur across the Kitsune dataset and CICIoT2023.

Reading the two tables together: Tenko's *per-attack* discrimination is in the same band on both per-attack datasets (mean AUC 0.956 on Kitsune, 0.883 on CICIoT2023, the gap driven almost entirely by the harder OS-Scan/recon and brute-force classes), and the matched attacks agree within ± 0.06 except OS Scan. The low 0.607 AUC–ROC on the Mateen combined-attack shift regime is *expected and on-message*: it is a deliberate stress test in which Tenko is designed to degrade conservatively rather than adapt, so it trails the fully adaptive drift baselines there while staying strong on the stationary per-attack evaluations.

7 Honest limitations and talking points for colleagues

- **"Tenko doesn't beat Kitsune everywhere on CICIoT2023."** Correct, and expected: the contribution is node aggregation for sustained/entity-conditioned attacks; that is exactly where it wins here (MITM, Mirai). On pure floods per-packet RMSE is already near-ceiling. We should frame CICIoT2023 as a *generalization* result, not a dominance claim.

- **"The published η values didn't work."** True — $\eta = 50/20$ is degenerate on this benign geometry. But this *supports* the paper's own framing of η as a per-deployment benign multiplier; we recalibrated using benign traffic only, no attack labels. Prepared answer: report the recalibration procedure explicitly.

- **"Your operating-point FPR is definitional."** Acknowledged in the report: we fit and measured FPR on the same benign slice. The out-of-sample test shows the FPR does not transfer ($\approx 0\%$ vs $\approx 30\%$). Prepared answer: headline threshold-independent AUC/EER; present the operating point clearly labelled as in-sample.

- **"Kitsune's FPR is 32% — unfair comparison."** The Kitsune baseline uses a Median+MAD threshold that lands at ≈ 0.3233 FPR; the two systems are therefore not at a matched FPR. We compare them on *threshold-independent* AUC/EER, which is immune to this.

- **"OS Scan looks much worse than in the paper."** Different traffic under the same name: CICIoT2023 reconnaissance overlaps benign more. Prepared answer: state this explicitly and lean on the AUC being a separability statement.

- **"It's only a pilot."** Yes — 150k benign lead + 30k benign test + 50k per attack, six classes. Scaling to more classes / more packets is future work; the pilot already exercises the full X1–X4 path on real per-packet source IPs.

- **Double-**tanh**.** Redundant in the released code; it barely moves AUC (≤ 0.011) but does affect the operating point. The main-paper numbers were generated under it, so they stand and are described accurately; new CICIoT2023 runs use the single-tanh normalization.

A Provenance of every key number

Every figure above is traceable to a committed artifact under `results/CICIoT2023/` or to a verified companion memo (`experiment_story.md`, `cross_dataset_comparison.md`, `manuscript_drafts.md`). Table 12 maps the load-bearing numbers to their source.

Compile note. This document is self-contained (no external figures or bibliography) and builds with `pdflatex cic_experiment_report.tex` (or `tectonic cic_experiment_report.tex`).

Table 12: Provenance map. CSV row numbers are 1-based including the header.

Claim	**Source file : location**	**Value**
Stream layout 150k/30k/50k = 230k; `benignLimit`= 150k	`stream_counts.csv:2-7`; `build_streams_ciciot2023.sh:21-23`	as tabled
benign_test = pkts 150,001–180,000, same trace	`build_streams_ciciot2023.sh:47-53`	30,000
tshark source-IP column 4 (IPv4) / 17 (IPv6)	`pcap2tsv_tenko.sh:4-9`	—
Committed η degenerate (TPR=FPR= 0, all classes)	`ciciot2023_committedEta_double.csv:3,5,7,9,11,13`	0/0
Recalibrated η @1% / @5%	`ciciot2023_metrics_recalibrated.csv:2,3`	40.699/2.854 ; 35.834/2.207
Double-tanh location	`example.py:157` + `results.py:805`	$\tanh(\tanh(\cdot))$
Single−double AUC ≤ 0.011 (Dict max +0.010911)	`ciciot2023_tanh_impact.csv:2-7`	≤ 0.011
Full-trace AUC/EER (Kitsune & Tenko)	`ciciot2023_per_class_metrics.csv: 2,3,6,7,10,11,14,15,18,19,22,23`	Table 5
Operating point @5% (in-sample)	`ciciot2023_per_class_metrics.csv:5,17,4,21,13,25`	Table 6
Kitsune constant FPR ≈ 0.3233	`ciciot2023_per_class_metrics.csv` (Kitsune `FPR` col)	0.323333
OOS held-out FPR 0% / $\approx 30\%$	`ciciot2023_metrics_recalibrated_oos.csv:2-5` (`heldout_benign_fpr`)	0.0 / 0.304 / 0.319
Paper Kitsune-dataset AUC/EER (9 attacks)	manuscript `tab:kitsune-tenko-new-highlighted`	Table 8
Matched-attack ΔAUC	`cross_dataset_comparison.md` (§2b)	Table 9
Raw corpus ≈ 548 GB; pilot target ≈ 100–150 MB	`TENKO_R2_REVISION_ROADMAP.md:78`	—